

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

1stSemester of M.Pharm (Quality Assurance) Examination

University Theory Examination January 2018

PHQA004 CLINICAL RESEARCH AND DRUG REGULATORY AFFAIRS

Date: 09/01/2018

Time: 10:00 A.M. – 01:00 P.M.

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1** Describe drug development process using diagram. What are Clinical Trials? Classify them. Differentiate various phases of clinical trials. **10**
- Q.2** (A) Describe in detail various types of Clinical Trial Design. **10**

OR

- Q.2** (A) What is BA-BE studies? Discuss its requirement in Drug development process. **05**
- (B) Discuss in brief about ICH quality guideline. **05**
- Q.3** Answer the following (Any three) **15**
- (A) Elaborate on ethical considerations during clinical trial protocol design.
- (B) Discuss Clinical research protocol in brief.
- (C) Write a note on Schedule Y in Brief.
- (D) How control group and population group are selected in clinical Trials?

SECTION-II

- Q.4** Write a Short note on NDA Approval Process. **10**
- Q.5** (A) What is DMF? Explain types and significance of DMF. **05**
- (B) Explain three Level of changes in site changes as per SUPAC guideline **05**

OR

- Q.5** (A) (i) Differentiate ANDA and NDA (ii) Concept of paragraph I to IV filing. **05**
- (B) Explain Format & Content of IND. **05**
- Q.6** Answer the following (Any three) **15**
- (A) Explain types of inspections carried out as per WHO guideline.
- (B) What is Medical Device? Classification of Medical Devices as per CDSCO with examples of devices.
- (C) Types of Application Procedures under MHRA.
- (D) Write a brief note on "Hatch Waxman Act".